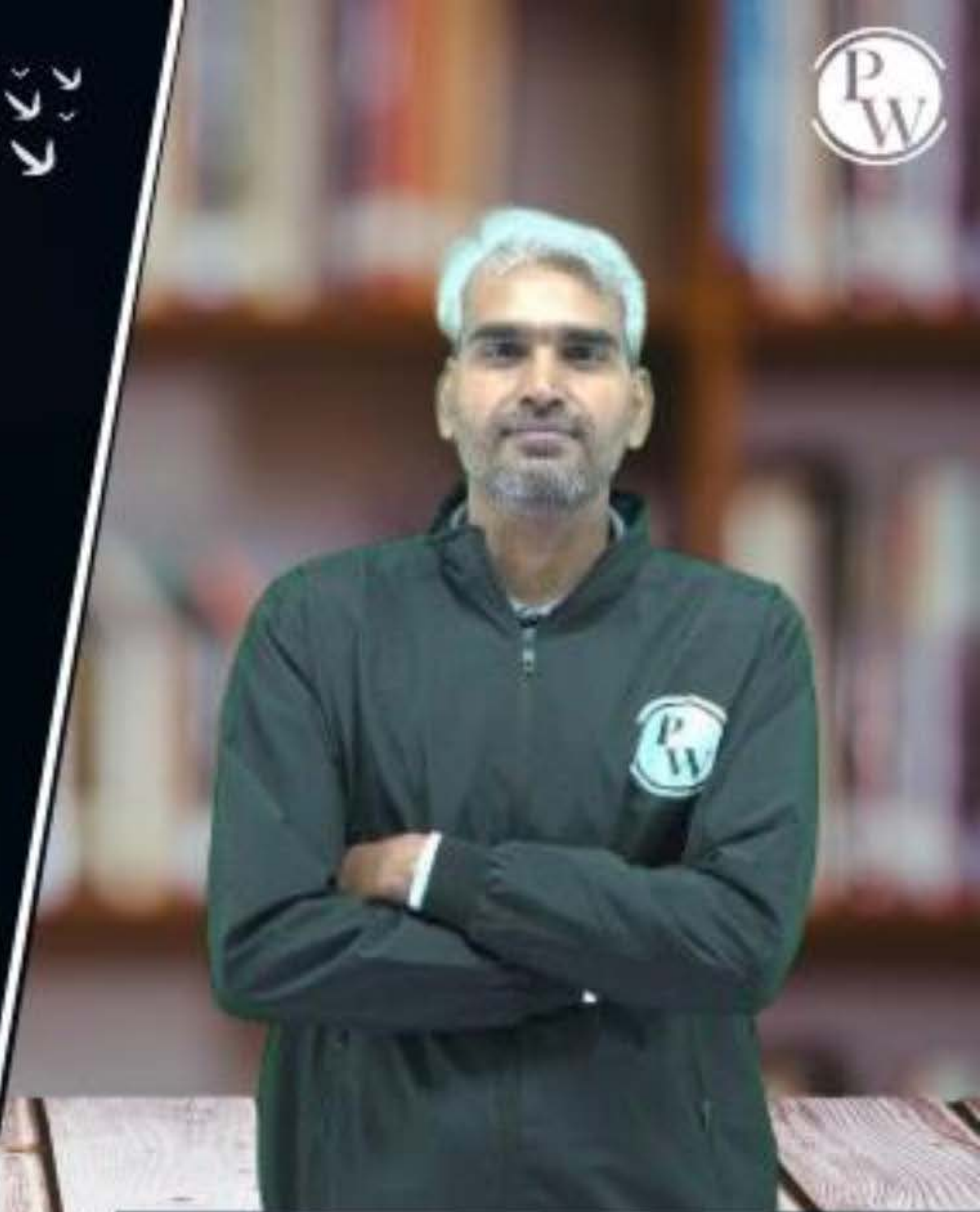
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture

Topic — Least upper bound (LUB)

Topic — Greatest lower bound (GLB)

Topic — Minimal and maximal elements in a POSET

Topic — Minimum (Least) element of POSET

Topic — Maximum (Greatest) element of POSET

Slide

Topics to be Covered

- Topic — Lattice
- Topic — Hasse diagram/POSET diagram
- Topic — Bounded lattice
- Topic — Complement of an element in a lattice
- Topic — Complemented lattice

Slide

$\boxed{\text{Lattice}}$

Note:-

In a POSET, lub and/or glb need not exist for every pair of elements of the set

Topic : Join Semi Lattice

* A POSET in which least upper bound exists for every pair of elements of the set is called Join-semi lattice

{ glb. may or may not exist for every pair of elements. }

Slide

Topic : Meet Semi Lattice

* A POSET in which greatest lower bound exists for every pair of elements of the set is called a Meet semi lattice.

{lub may or may not exist for every pair of elements}

Slide

- A POSET in which both least upper bound (lub) as well as greatest lower bound (glb) exists for every pair of elements of the set is called a lattice

* A POSET that is a Join semi lattice as well as a meet semi lattice is called a lattice

eg ① $A_1 = \{2, 3, 4, 6, 15\}$ and $(A_1, \div)$ is a POSET

glb = DNE ∴ Not a Meet semi lattice (for 2, 3)

lub = DNE ∴ Not a Join semi lattice (for 6, 15)

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

② $A_2 = \{2, 3, 4, 6, 12\}$ and $(A_2, \div)$ is a POSET

glb = DNE ∴ Not a Meet semi lattice (for 2, 3)

lub exist for Every pair of elements, ∴ A Join semi lattice

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

③ $A_3 = \{1, 2, 3, 4, 6, 15\}$ and $(A_3, \div)$ is a POSET

glb exist for Every pair ∴ A Meet semi lattice

lub = DNE ∴ Not a Join semi lattice (for 6, 15)

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

④ $A_4 = \{1, 2, 3, 4, 6, 12\}$ and $(A_4, \div)$ is a POSET

Both lub as well as glb exist for Every Pair, ∴ it is a lattice

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

Note :-

1. If D_n is the set of all positive divisors of a positive integer 'n', then $(D_n, \div)$ is always a lattice.

$D_{12} = \{1, 2, 3, 4, 6, 12\}$
$D_{21} = \{1, 3, 7, 21\}$

2. If 'A' is any finite set, and $P(A)$ is the power set of set A, then $(P(A), \subseteq)$ is always a lattice

3. If A is any set of real numbers, then $(A, \leq)$ is always a lattice

Topic : Lattice

A lattice is denoted by $[L, \vee, \wedge]$

lub ($\vee$)

glb ($\wedge$)

L → Name of lattice

Slide

Properties w.r.t. Lattice

- Let $[L, \vee, \wedge]$ be a lattice then following properties will always hold true in a lattice

1. Idempotent Law: (i) $a \vee a = a$, $\forall a \in L$
   (ii) $a \wedge a = a$, $\forall a \in L$
2. Commutative Law (i) $a \vee b = b \vee a$, $\forall a, b \in L$
   (ii) $a \wedge b = b \wedge a$, $\forall a, b \in L$
3. Associativity law (i) $(a \vee b) \vee c = a \vee (b \vee c)$, $\forall a, b, c \in L$
   (ii) $(a \wedge b) \wedge c = a \wedge (b \wedge c)$, $\forall a, b, c \in L$
4. Absorption Law (i) $a \vee (a \wedge b) = a$, $\forall a, b \in L$
   (ii) $a \wedge (a \vee b) = a$, $\forall a, b \in L$

These four properties will always hold true in any lattice

Note:- Distributive Property need not hold true for all lattices

i.e. (i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c), \ \forall a, b, c \in L$

and (ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c), \ \forall a, b, c \in L$

If above distributive property holds true in a lattice, then that lattice is called a distributive lattice

If distributive property does not hold true for at least one combination of a, b & c, then lattice is not a distributive lattice

* **Note :→**

Important

In a lattice least upper bound as well as greatest lower bound must exist for every Pair of elements and they must be unique

☆

$\boxed{\text{Hasse Diagram}}$

↳ Constructed for POSET

∴ Also known as POSET diagram

Topic : Hasse Diagram / POSET Diagram

In a Hasse diagram of a $\boxed{\text{POSET}}$,

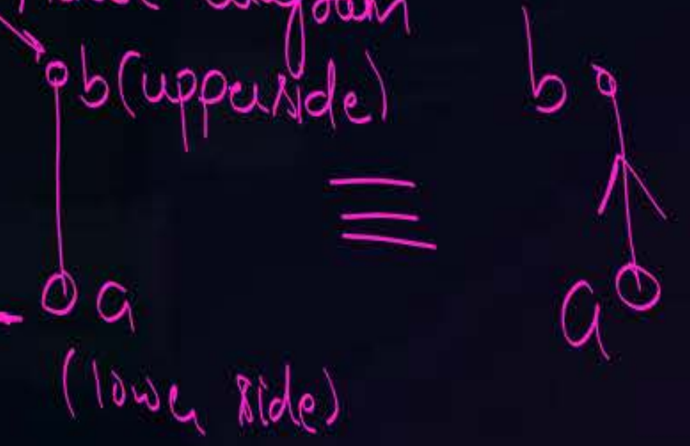
Reflexive
Anti-symmetric
Transitive

1. There is a vertex corresponding to every element of set.

2. There is an edge from vertex a to vertex b if and only if a relates to b and there is no element x in the set such that a relates to x, and x relates to b.(Transitivity is implied in the Hasse diagram not represented explicitly)

3. No self-loop on the vertices (i.e. reflexivity is implied in the Hasse diagram not represented explicitly).

i.e. if $(a,b) \in R$ then in Hasse diagram



Generally

4. It is not directed but it uses implied upward orientation.

{But some authors defines it as a directed graph}

Slide

Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSETs

1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$
2) $(D_6, \div)$
3) $(D_{12}, \div)$
4) $(\{2, 3, 4, 6\}, \div)$
5) $(\{2, 3, 6, 12\}, \div)$
6) $(\{1, 2, 3, 4, 6, 9\}, \div)$

Slide

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$

$$\leq \; = \left\{ \begin{array}{lllll} (-1,-1) & (-1,0) & (-1,2.5) & (-1,4) & (-1,6) \\ & (0,0) & (0,2.5) & (0,4) & (0,6) \\ & & (2.5,2.5) & (2.5,4) & (2.5,6) \\ & & & (4,4) & (4,6) \\ & & & & (6,6) \end{array} \right\}$$

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$ ← This POSET is also a TOSET

Whenever we construct the Hasse diagram for a TOSET it will look like a linear chain

∴ Totally order Set (TOSET) are also known as linearly ordered set

$$\leq = \left\{ \begin{matrix} (-1,-1) & (-1,0) & (-1,2.5) & (-1,4) & (-1,6) \\ & (0,0) & (0,2.5) & (0,4) & (0,6) \\ & & (2.5,2.5) & (2.5,4) & (2.5,6) \\ & & & (4,4) & (4,6) \\ & & & & (6,6) \end{matrix} \right\}$$

Q:- Let $A = \{a, b, c, d\}$

How many Total order relation can be defined on set A.

Hasse diagram of a total order relⁿ will look like a linear chain

'and'

'4' elements of the set can be arranged at '4' vertices of Hasse diagram in $4!$ ways

$\therefore$ $4!$ Total order relation can be defined on set A.

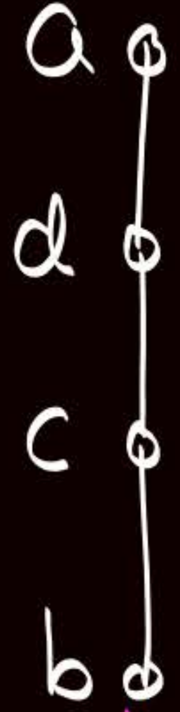
Ans $= 4! = 24$

Note:- If A is a set with 'n' elements, then $n!$ different total order relation are possible on set A.

Let $A = \{a, b, c, d\}$

and assume two Hasse diagrams

&



Let R_1 is the Total order rel^n w.r.t. Hasse diagram

$R_1 = \{(a,a)\ (a,b)\ (a,c)\ (a,d)$
$(b,b)\ (b,c)\ (b,d)$
$(c,c)\ (c,d)$
$(d,d)\}$

Let R_2 is the Total order rel^n w.r.t. Hasse diagram

$R_2 = \{(b,b)\ (b,c)\ (b,d)\ (b,a)$
$(c,c)\ (c,d)\ (c,a)$
$(d,d)\ (d,a)$
$(a,a)\}$

Hasse diagram is different
So. Corresponding Total order rel^n are also different

Note:- If POSET are different, then Hasse diagram will also be different on same set

eg: Let $A = \{a, b, c, d\}$

(i) $R_1 = \{(a,a), (b,b), (c,c), (d,d)\}$

└ (A, R_1) is a POSET

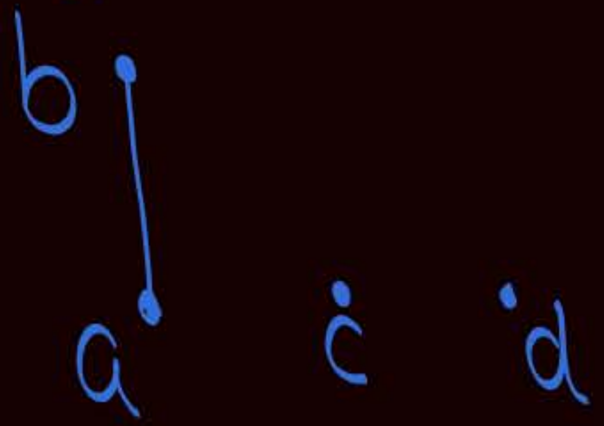
Hasse diagram will be

if elements are not Comparable then they may be at the same level

(ii) $R_2 = \{(a,a), (b,b), (c,c), (d,d), (a,b)\}$

└ (A, R_2) is a POSET

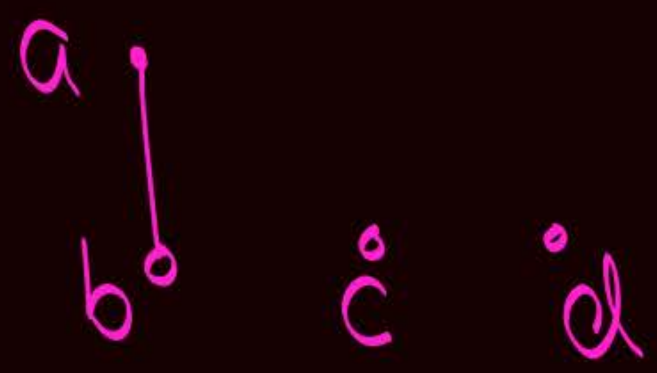
Hasse diagram will be



(iii) $R_3 = \{(a,a), (b,b), (c,c), (d,d), (b,a)\}$

└ (A, R_3) is a POSET

Hasse diagram will be



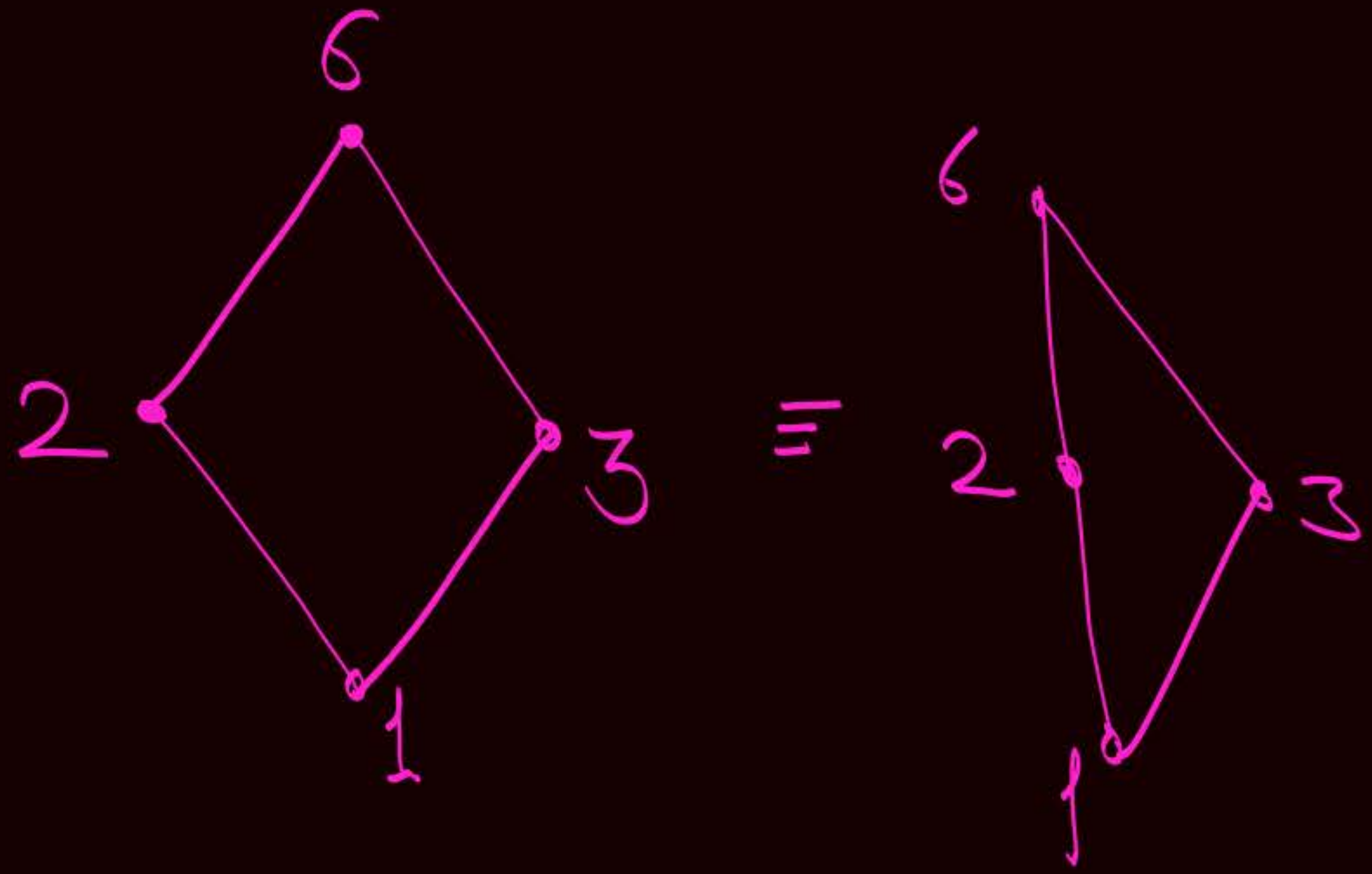
Q (ii) Construct the Hasse diagram for POSET,

$(D_6, \div)$

$\downarrow$

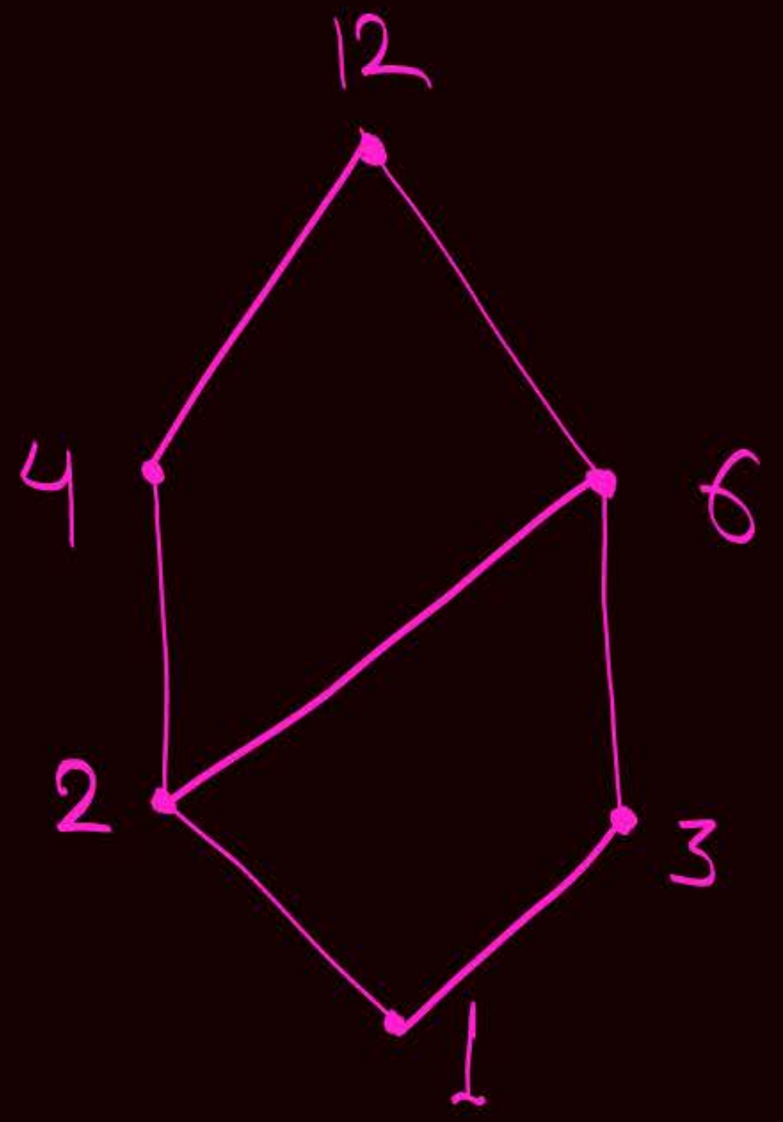
$D_6 = \{1, 2, 3, 6\}$

i.e., $(\{1, 2, 3, 6\}, \div)$

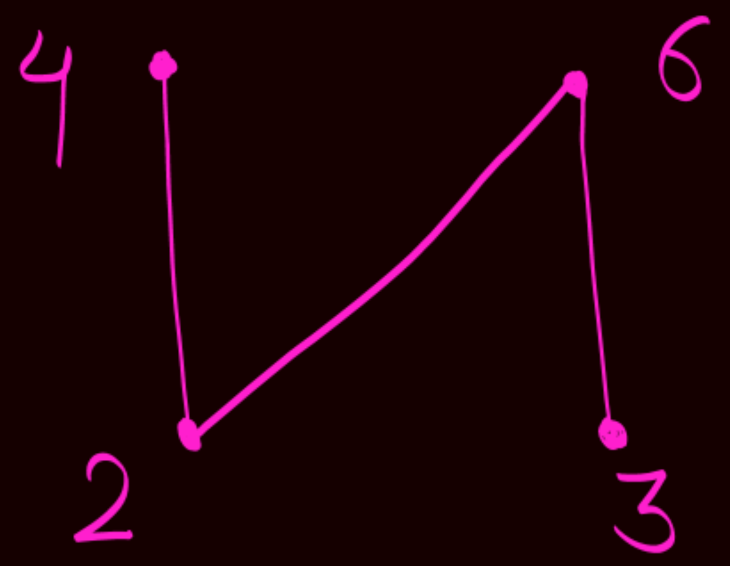


Q (iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$

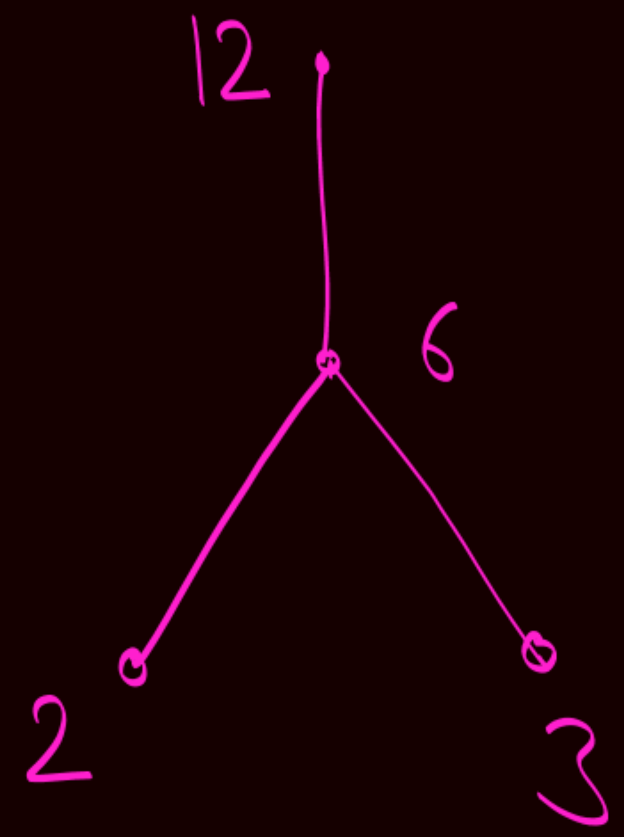


Q (iv) Construct the Hasse diagram for POSET, $(\{2,3,4,6\}, \div)$



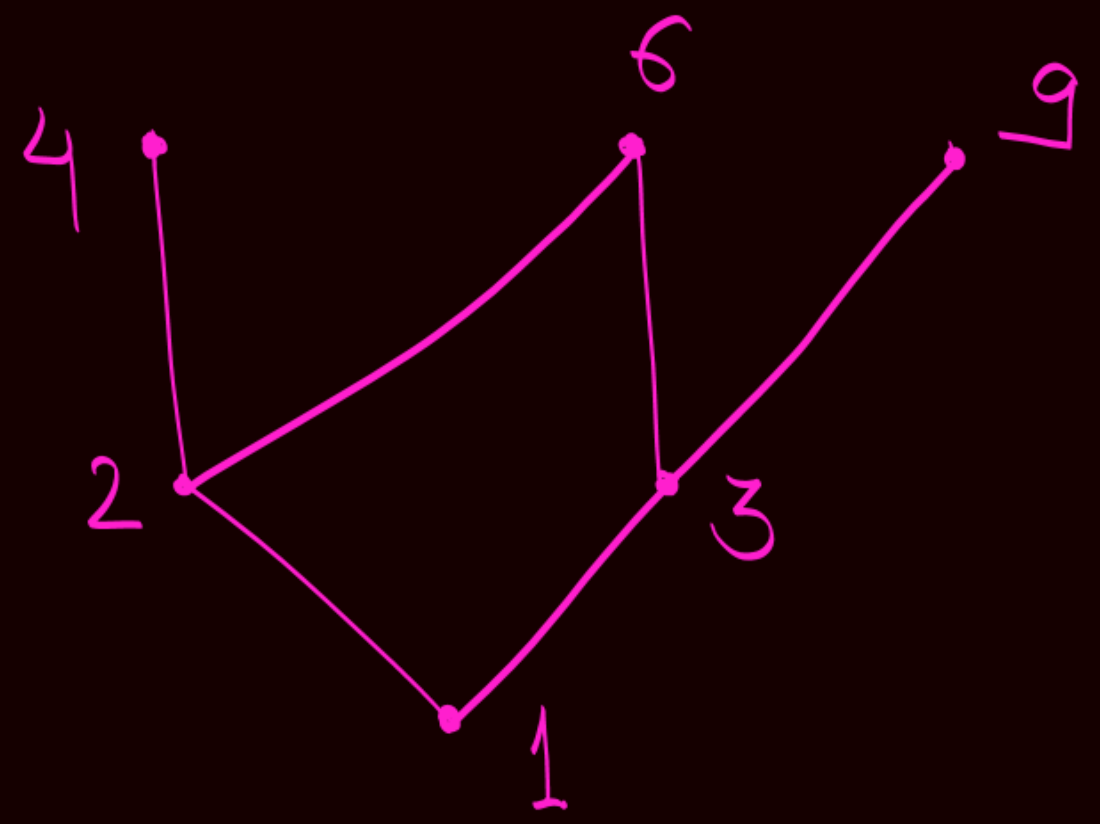
Q (V) Construct the Hasse diagram for POSET,

$(\{2, 3, 6, 12\}, \div)$



Q (vi) Construct the Hasse diagram for POSET,

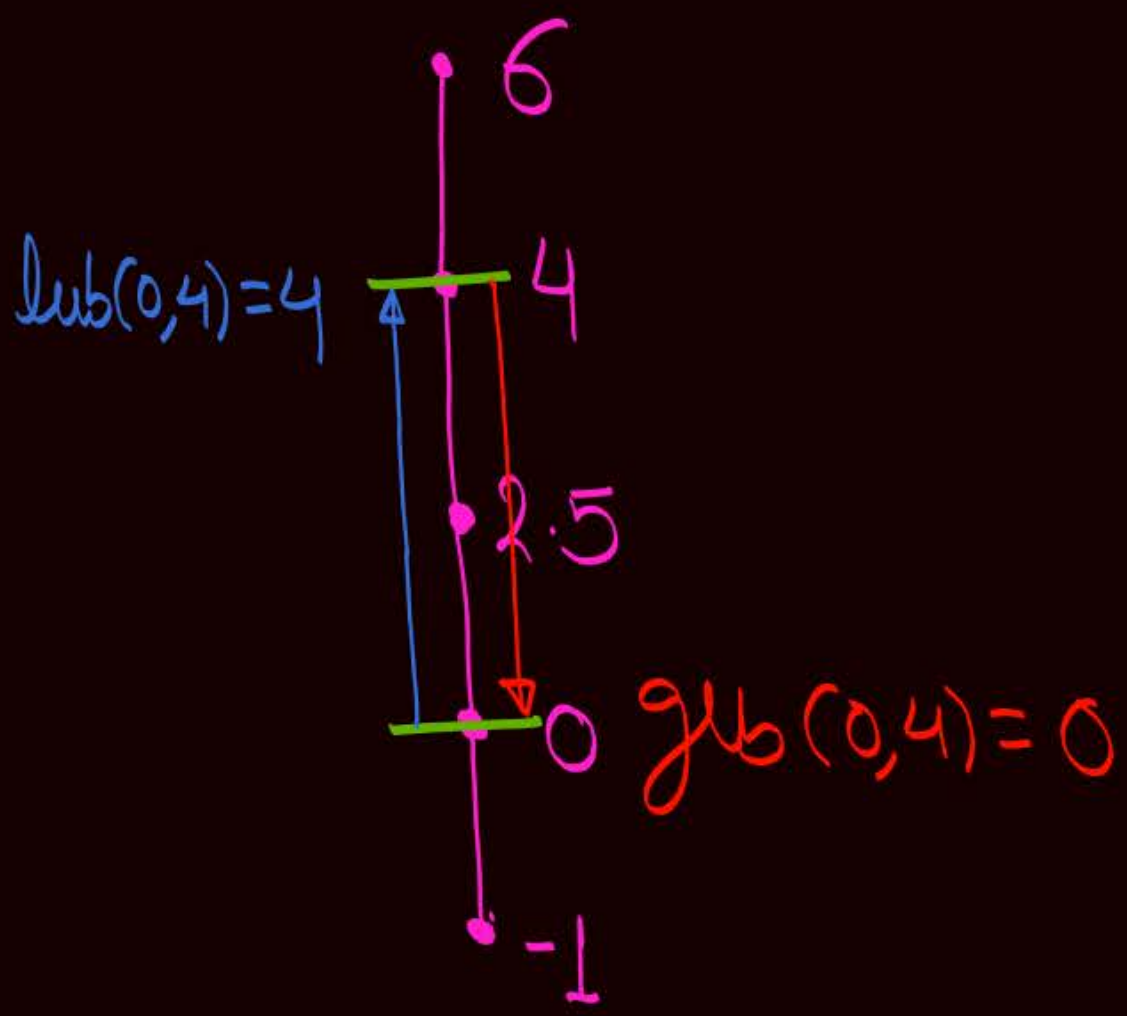
$$(\{1, 2, 3, 4, 6, 9\}, \div)$$



Understand lub & glb w.r.t. Hasse diagram

(1) $(\{-1, 0, 2.5, 4, 6\}, \leqslant)$ less than or equal

Hasse diagram for given POSET.



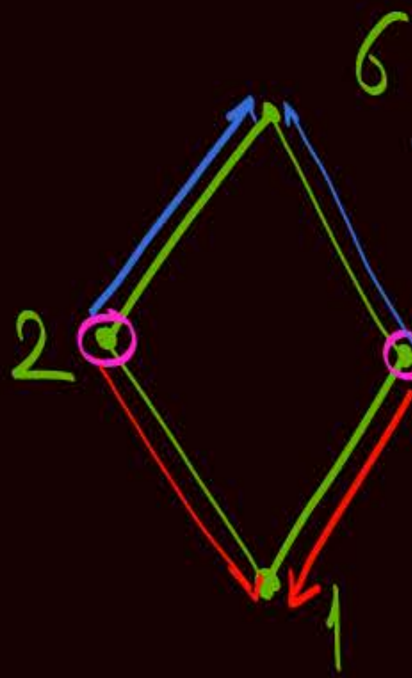
If we choose two elements on the same line, then the element at the upper side on the line is lub, and the element at the lower side on the line is glb

+ In the given POSET lub as well as glb exist for every pair of elements of the set, Hence POSET is a lattice

Note:-

$\boxed{\text{Every TOSET is always a lattice but converse of the statement need not be true}}$

Q (ii) Construct the Hasse diagram for POSET, $(D_6, \div)$



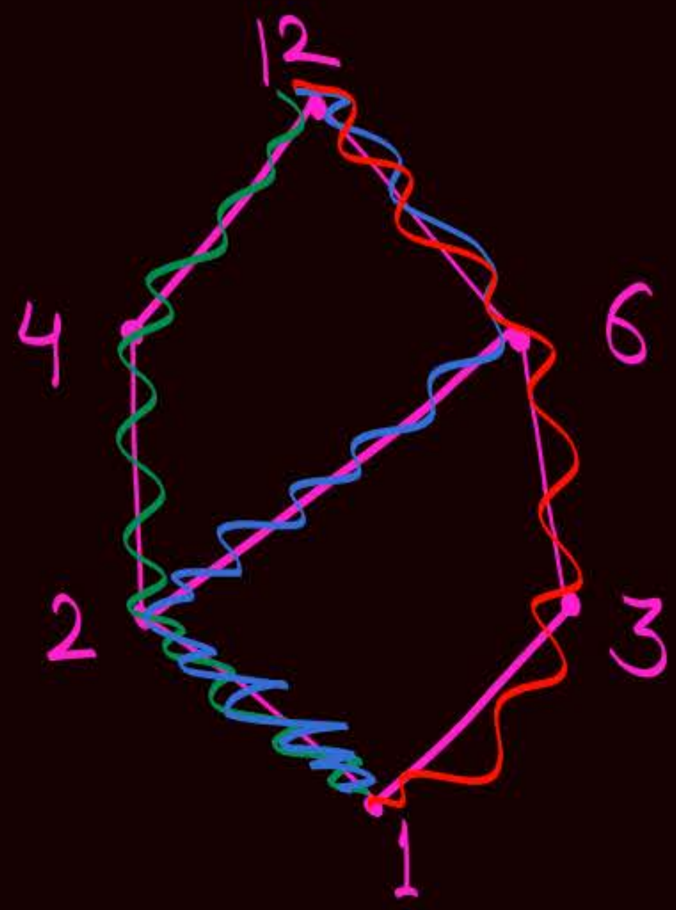
lub(2,3) = 6

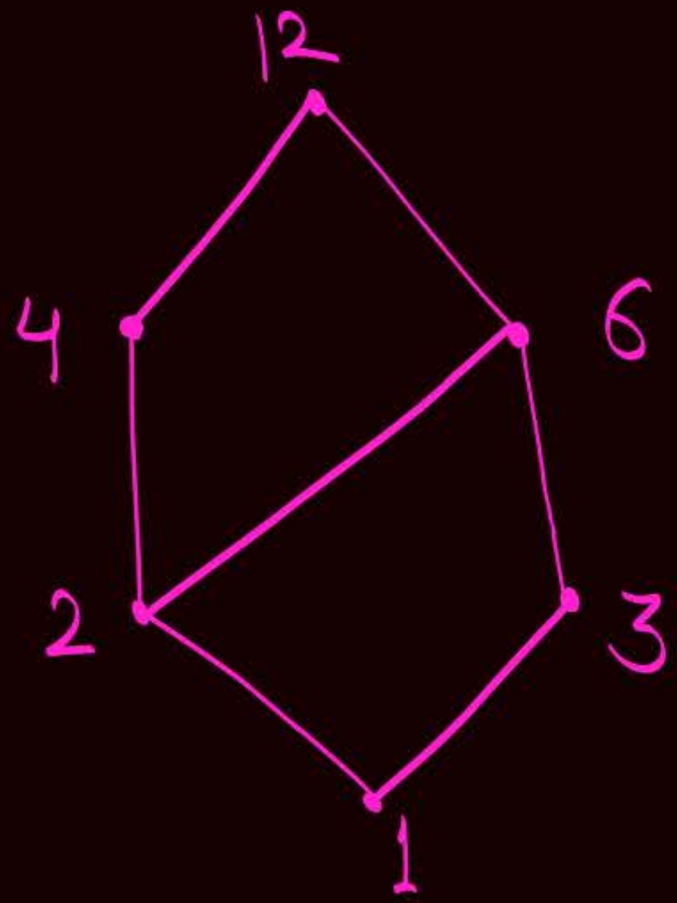
glb(2,3) = 1

Lub as well as glb exist for every pair of elements of the set

$\therefore$ Given POSET is a lattice

Q (iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$





$$\begin{rcases}
\text{lub}(2,3) = 6 \\
\text{glb}(2,3) = 1 \\
\text{lub}(3,4) = 12 \\
\text{glb}(3,4) = 1 \\
\text{lub}(4,6) = 12 \\
\text{glb}(4,6) = 2
\end{rcases}$$

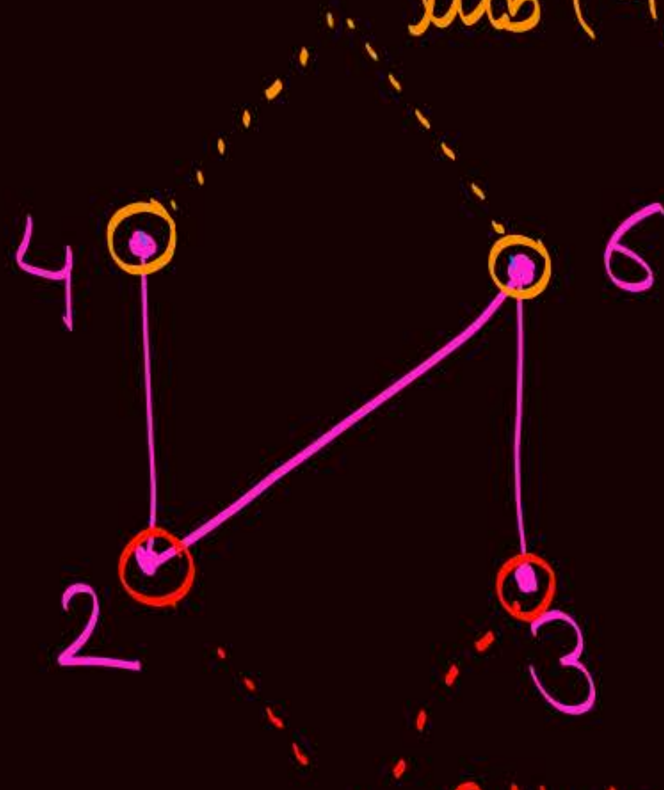
Lub as well as Glb exist for Every Pair of elements of POSET

$\therefore$ Given POSET is a lattice

Q (iv) Construct the Hasse diagram for POSET,

$(\{2, 3, 4, 6\}, \div)$

lub(4,6) = does not exist



∴ POSET is not a Join semi lattice

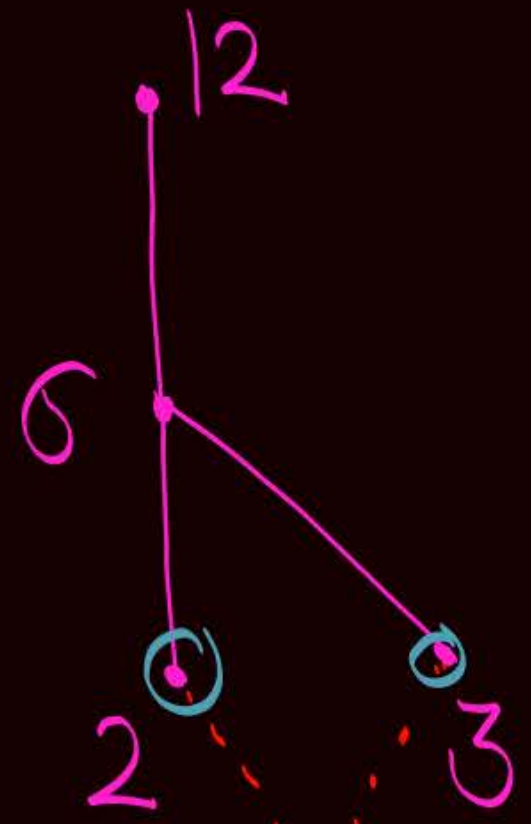
∴ POSET is not a meet semi lattice

glb(2,3) = does not exist

Neither a Join semi lattice, nor a Meet semi lattice

Q (V) Construct the Hasse diagram for POSET,

$(\{2, 3, 6, 12\}, \div)$



$lub(2,3) = 6$

↳ lub exist for Every Pair of elements of the set.

∴ POSET is a Join semi lattice

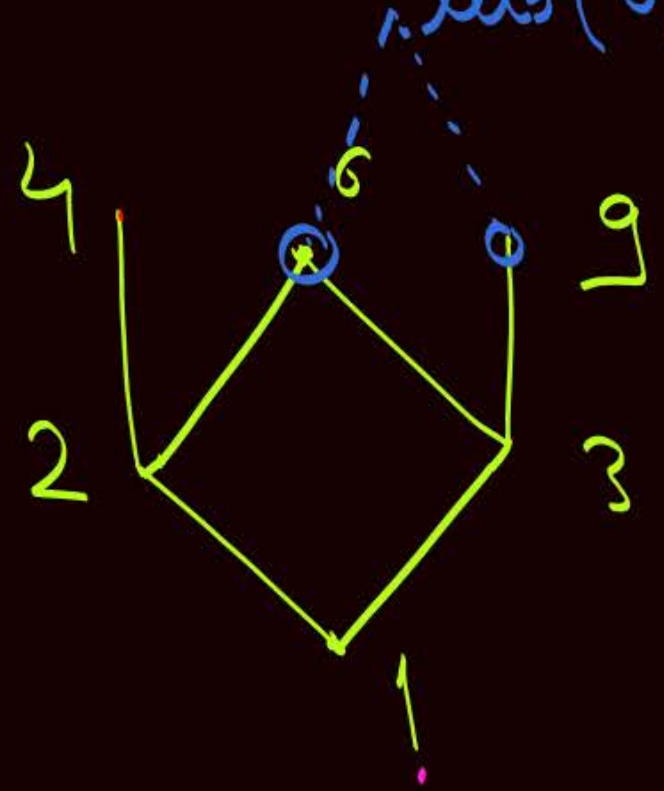
$glb(2,3)$ = does not exist

→ ∴ POSET is not a Meet semi lattice ⇒ ∴ Not a lattice

Q (vi) Construct the Hasse diagram for POSET,

$(\{1, 2, 3, 4, 6, 9\}, \div)$

lub(6, 9) = does not exis



∴ POSET is not a Join semi lattice ⇒ Hence not a lattice

glb exist for Every pair of elements of the set

∴ POSET is a Meet semi lattice

Maximal & Minimal elements

+

Maximum & Minimum element

w.r.t Hasse diagram

(1) $(\{-1, 0, 2.5, 4, 6\}, \leqslant)$

Maximal element
↳ (Unique maximal)
↳ ∴ Maximum element as well

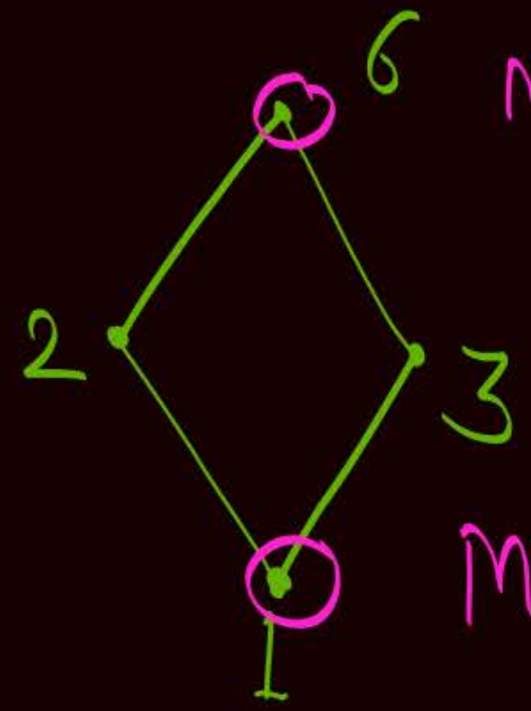
Minimal element
↳ (Unique Minimal)
↳ ∴ Minimum element as well

Q (ii) Construct the Hasse diagram for POSET,

$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$

i.e, $(\{1, 2, 3, 6\}, \div)$

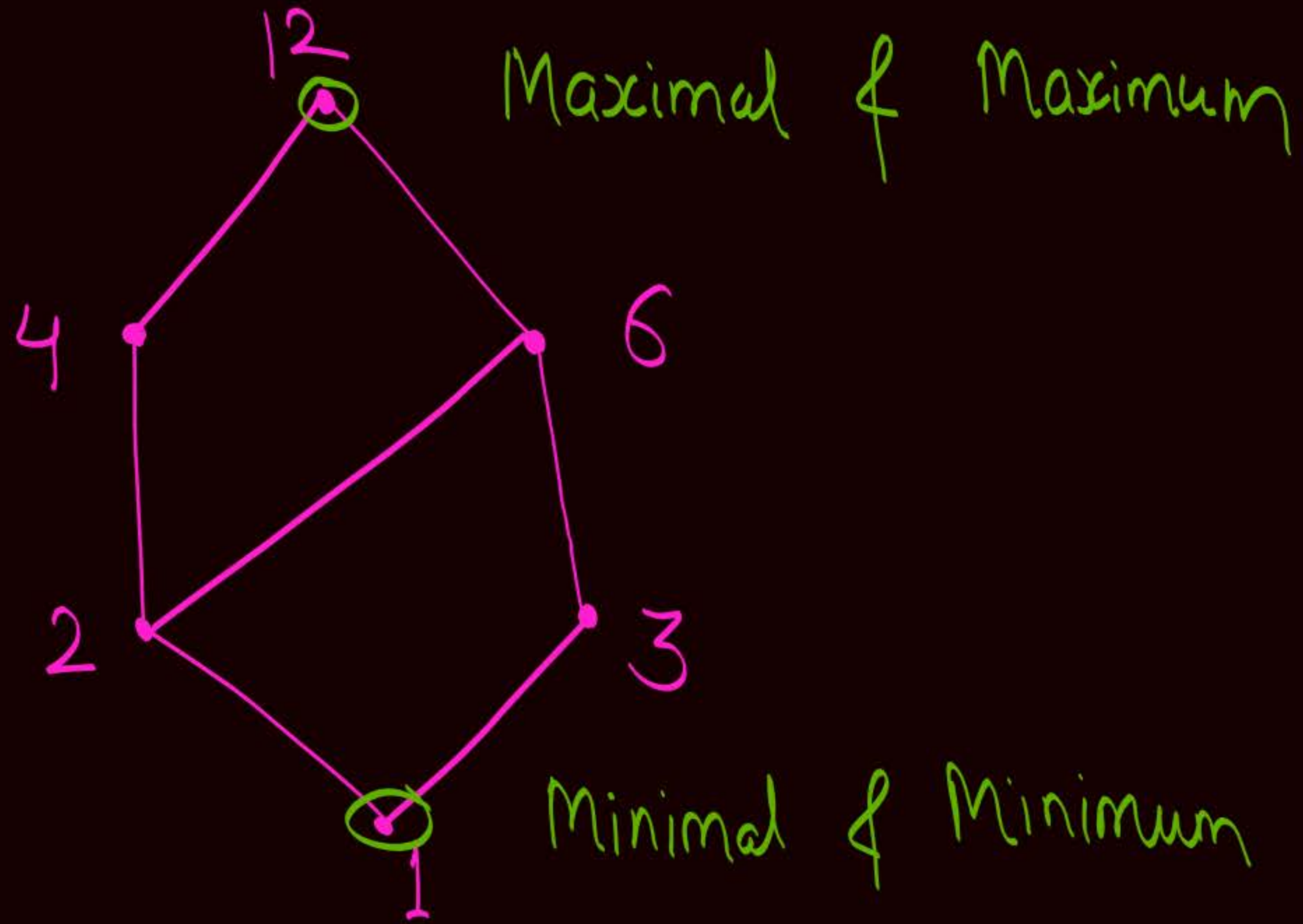


Maximal & Maximum

Minimal & Minimum

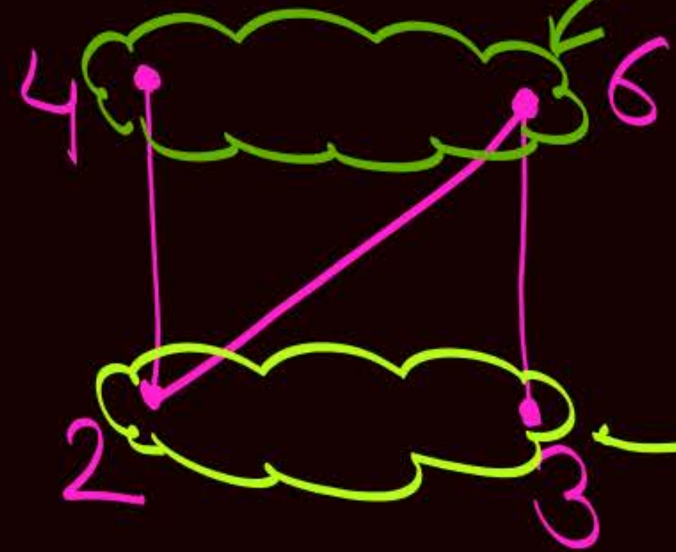
Q (iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



Q (iv) Construct the Hasse diagram for POSET,

$(\{2, 3, 4, 6\}, \div)$



Both are maximal

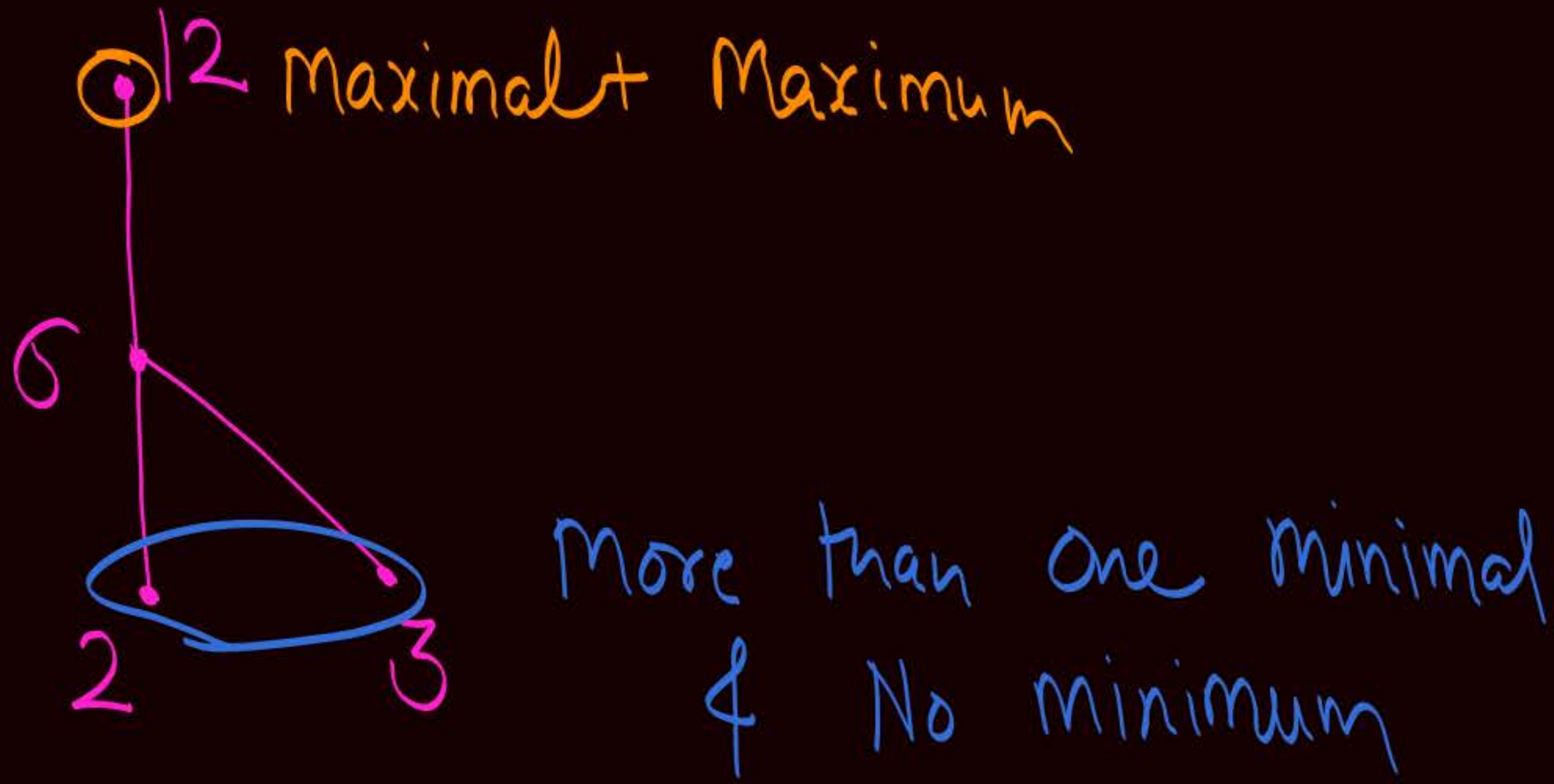
{More than one maximal ∴ No maximum}

Both are Minimal

{More than one Minimal ∴ No Minimum}

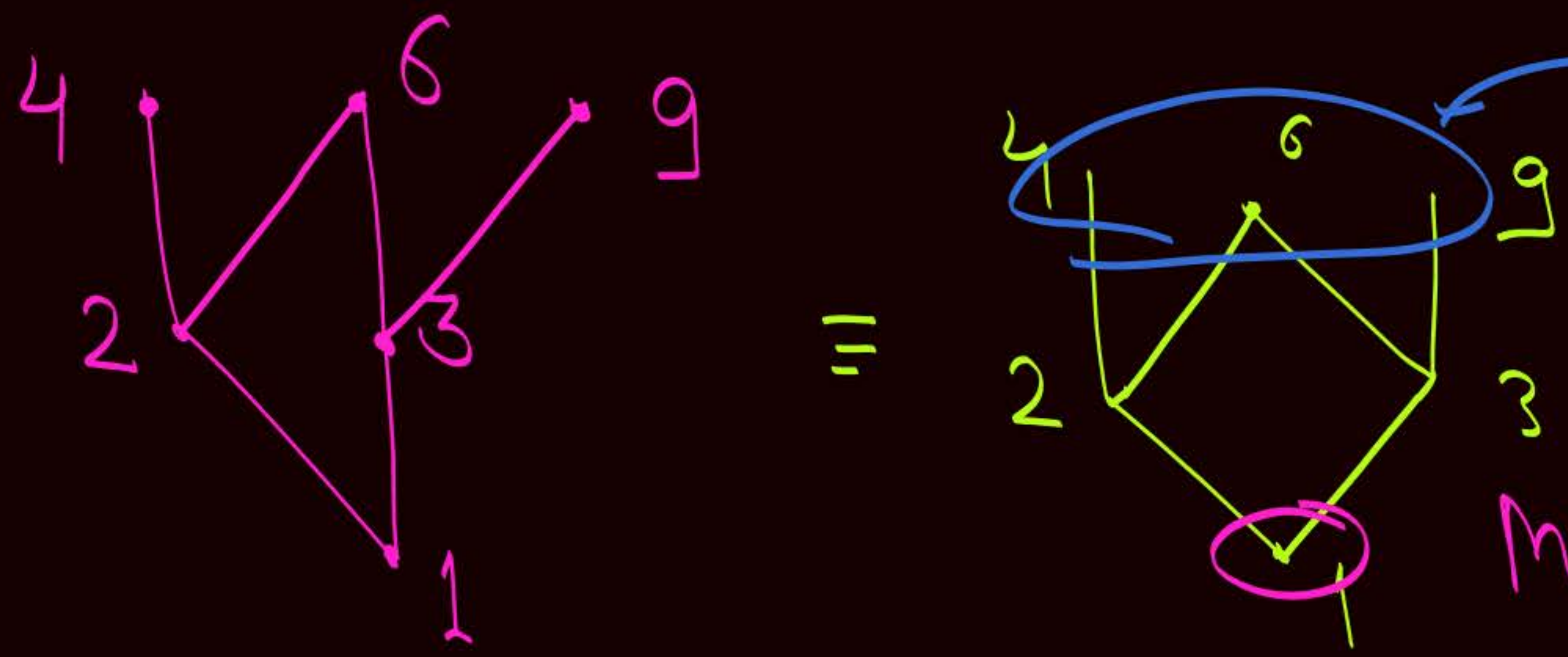
Q (V) Construct the Hasse diagram for POSET,

$(\{2, 3, 6, 12\}, \div)$



Q (vi) Construct the Hasse diagram for POSET,

$(\{1, 2, 3, 4, 6, 9\}, \div)$



More than one Maximal

∴ No maximum

Minimal + Minimum

Note:-

1. In a POSET if there exist two or more maximal elements then it can not be a Join semi lattice { ∴ Not a lattice }

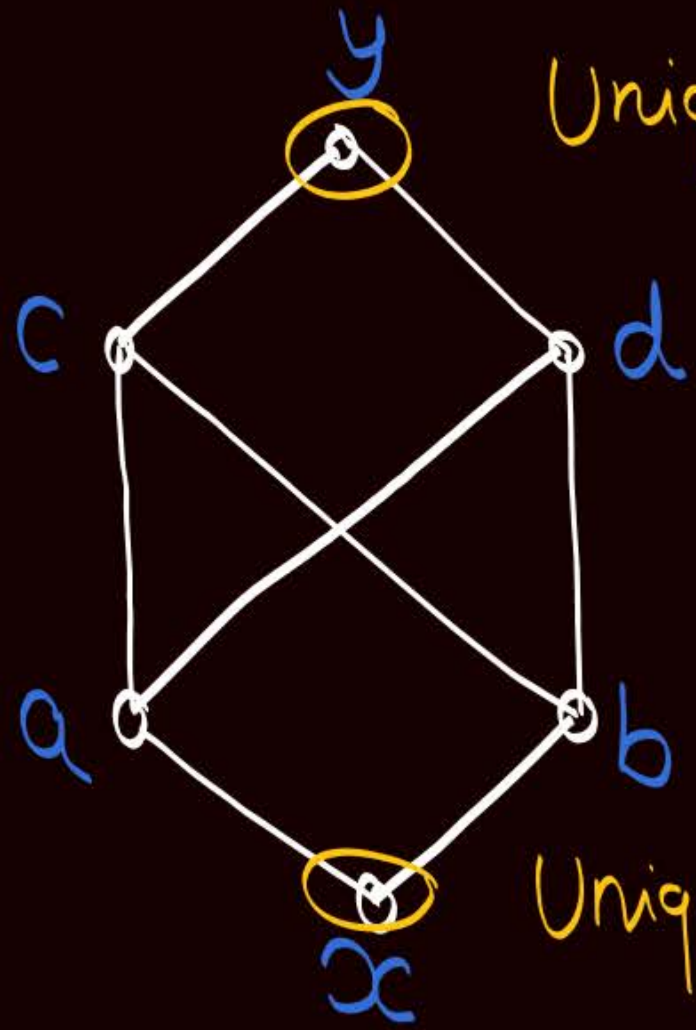
2. In a POSET if there exist two or more minimal elements then it can not be a Meet semi lattice { ∴ Not a lattice }

* **Note** :→

Important

In a lattice least upper bound as well as greatest lower bound must exist for every Pair of elements and they must be unique eg:→ Consider the following POSET diagram

Check whether the POSET corresponding to given POSET diagram is a lattice or not?



Unique maximal

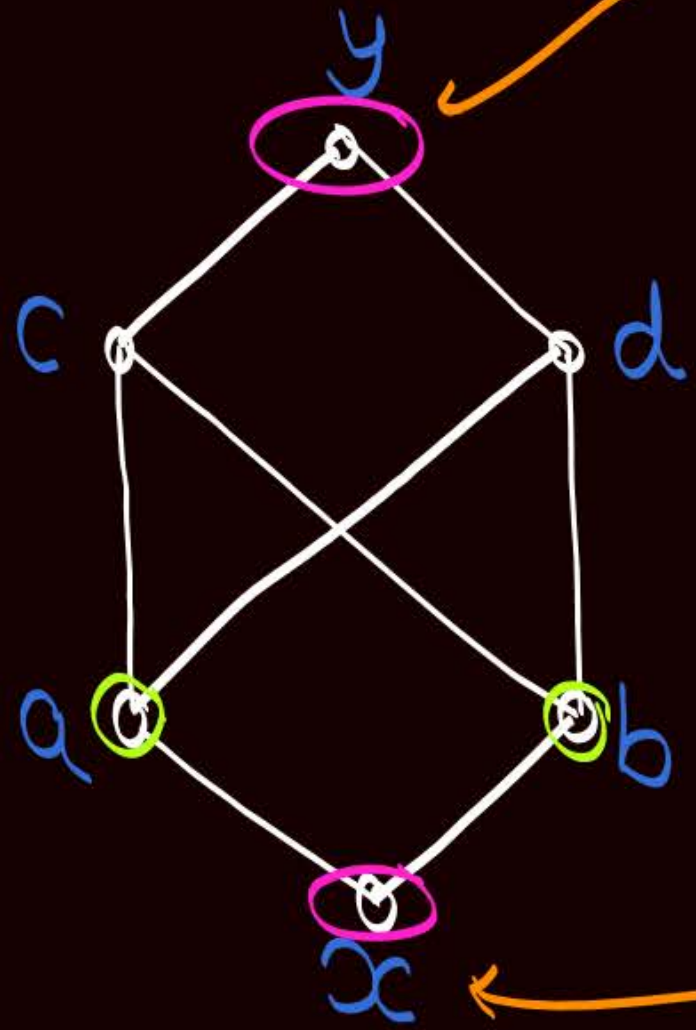
∴ Maximum as well

Unique minimal

∴ Minimum as well

eg: → Consider the following POSET diagram
Check whether the POSET corresponding to given POSET diagram is a lattice or not?

'y' is the unique maximal element
∴ y is the maximum element as well



There exist maximum as well as minimum element in the POSET.
but lub(a,b) = Either does not exist (Or) (Both c & d are lub)
& glb(c,d) = Does not exist (Or) (Both a & b are glb)
In a lattice lub as well as glb must exist and they must be unique ∴ Not a lattice

x is the unique minimal element
∴ x is the minimum element as well

→ There exists some POSET, in which both minimum as well as maximum element exist but still they are not lattices.

Q: If POSET is a lattice, then there exist both minimum as well as maximum elements? True or False

This statement is False

Because lattice may be w.r.t. infinite set.

examples on next slide

Note:→ It is not necessary for a lattice to have maximum and/or minimum element, because the set may be an infinite set

$(N, \leq)$ is a lattice,

Maximum element : Does not exist

Minimum element $= 1$

eg: $(Z, \leq)$ is a POSET

Hasse diagram

* lub exist for every Pair of elements as well as glb exist for every Pair of elements

Hence it is a lattice,

But, maximum element does not exist

And Minimum element also does not exist.

* Minimum and Maximum element can exist even if the set corresponding to the POSET is an infinite set

eg: $A = \{x \mid \underbrace{x \in R \text{ and } 0 \leq x \leq 1}_{\text{it is an infinite set}}\}$

$(A, \leq)$ is a POSET

Minimum element is '0'

Maximum element is '1'

→ <u>Note:</u>

$\boxed{\text{If lattice is a finite lattice, then both Minimum \& maximum elements will exist}}$

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

And let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,2), (4,5)\}$ is a Partial order on set A.

How many total order relation are possible on set A such that relation 'R' is included in each total order relation

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

How many partial order relation are possible on set A, such that the cardinality of each partial order relation is exactly '7'

$\boxed{\text{Bounded Lattice}}$

→ A lattice in which both minimum & maximum elements are present is called a bounded lattice

Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

① If there exists an element $I \in L$
s.t. $a \vee I = I \quad \forall a \in L$.
then 'I' is called Universal upper bound
(or) Upper bound of lattice L

"I" is actually the maximum element of lattice.

② If there exist an element $O \in L$.
such that, $a \wedge O = O, \ \forall a \in L$
then, 'O' is called universal lower bound
(or) lower bound of lattice L

'O' is actually the minimum element

* A lattice $[L, \vee, \wedge]$ is called a bounded lattice if and only if both 'I' and 'O' exist in the lattice

(or)

* A lattice is called a bounded lattice if and only if both maximum & minimum element exist.

Slide

Note :- ① Every finite lattice is a bounded lattice

② If lattice is an infinite lattice then it may or may not be a bounded lattice

eg: $(Z, \leq)$ and $(N, \leq)$ are example of unbounded lattice where both Z & N are infinite sets

Bounded lattice w.r.t. infinite set

eg: let $A = \{x \mid x \in R \text{ and } 0 \leq x \leq 1\}$

→ A is an infinite set.

We know $(A, \leq)$ is a lattice where 'A' is an infinite set

but 0 = Minimum element

& 1 = Maximum element

→ Both minimum & maximum element exist

∴ it is a bounded lattice.

Topic : Complements of an element in a lattice

* Complements of an element of lattice $[L, \vee, \wedge]$ Can exist only if lattice 'L' is a bounded lattice

P only if Q

$\equiv$

If P then Q

$\equiv$

$P \rightarrow Q$

→ (i) If Complement exist for an element of lattice $[L, \vee, \wedge]$ then L is a bounded lattice

(ii) If Complement does not exist for an element of lattice $[L, \vee, \wedge]$, then 'L' may or may not be bounded

(iii) if $[L, \vee, \wedge]$ is not a bounded lattice, then Complement can never exist for any element of lattice L

Slide

Topic : Complement of an element in a lattice

Let $[L, \vee, \wedge]$ be a bounded lattice,

Where, I = Universal upper bound of lattice L {i.e. Maximum element}

& 0 = Universal lower bound of lattice L {i.e. Minimum element}

* For an element $a \in L$

if there exist some elements 'b' in set L

such that (i) $a \vee b = I$ {Maximum} ⇒ i.e. lub(a,b) = Maximum element

& (ii) $a \wedge b = 0$ {Minimum} ⇒ i.e. glb(a,b) = minimum element

then, a & b are called Complement of each other w.r.t lattice $[L, \vee, \wedge]$

Slide

Note:

① In a bounded lattice,
'I' and 'O' are always Complement of each other and they do not have any other Complement.

i.e., The only Complement of Maximum element (I) is minimum element (O)
and The only Complement of minimum element (O) is maximum element (I)

Imp.

② In a bounded lattice,
Complement need not exist for every element of the lattice, And if exists it need not be unique
{i.e. some elements may have more than one Complement

→ In a bounded lattice number of Complements of an element of the lattice may be 'O' or '1' or more

Topic : Complemented lattice

A lattice is called a Complemented lattice if and only if at least one Complement exist for Every element of the lattice

* In a Complemented lattice every element of the lattice will have 1 or more Complements

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2 mins Summary

- **Topic** Lattice
- **Topic** Hasse diagram/POSET diagram
- **Topic** Bounded lattice
- **Topic** Complement of an element in a lattice
- **Topic** Complemented lattice

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THANK - YOU

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